UCLA Anderson School of Management
Master of Science in Business Analytics

2026 Python Fundamentals Workshop
(DRAFT) Syllabus and LabAssist

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Course Description

Goal: Maximize your Python learning experience to prepare you to use Python effectively in the MSBA program, summer internships and full-time jobs.

- I'll guide you through a hands-on code-intensive presentation of Python - the world's most popular programming language. Using the interactive IPython interpreter and Jupyter Notebooks, we'll work through code examples ranging from individual code snippets to large scripts.
- I'll introduce popular free open-source libraries used in AI and data science, including NumPy for high-performance array computing, Pandas for powerful data wrangling, Matplotlib and Seaborn for visualization, and scikit-learn for AI machine learning.
- You'll become familiar with Python, its popular programming idioms, key timesaving Python Standard Library modules and additional open-source libraries.
- Through the Intro to Data Science sections at the ends of Chapters 5-9, you'll program in Python doing simulations with static and dynamic visualizations, Pandas Series and DataFrames, data wrangling and CSV files.
- On the AI side, in lecture 7, I've traditionally covered machine learning with scikit-learn. By the time we get there this year, you'll already be in your MSBA machine learning class. So, this year, I am changing the final lecture to generative and agentic AI. I'll demo programming with the OpenAI Python SDK, the OpenAI Agents SDK, and Codex for vibe coding and agentic engineering.

Key Topics in the Course

- Interactive Python coding in IPython and Jupyter Notebooks.
- Python code snippets and larger scripts.
- Built-in types, strings, input/output.
- Control statements, functions.
- Functional-style programming: Filter/map/reduce, lambdas, comprehensions.
- Collections: Lists, tuples, dictionaries, sets, high-performance NumPy arrays, Pandas Series and Pandas DataFrames.
- Static and dynamic 2D and 3D visualizations with Matplotlib and Seaborn.
- Text files, JSON serialization, CSV files, exception handling.
- Intro to object-oriented programming: Custom classes, inheritance, and polymorphism with duck typing.
- Demos of key data science and AI libraries that enable you to create powerful applications with minimal code.
- Generative AI programming with the OpenAI Python SDK.
- Agentic AI programming with the OpenAI Agents SDK.
- Vibe Coding and agentic engineering with OpenAI Codex.

Generative AI (GenAI)

- Throughout the workshop, I'll discuss and various generative AIs (genAIs) and how you can use them to enhance your learning experience.
- The models you'll create in industry will be used by companies to make critical decisions. GenAI is a powerful tool in your toolbox, but you cannot use genAI code in business-critical models without first proving the code is correct. **And you cannot prove code is correct unless you understand it.**
- **GenAIs hallucinate** - that is, they make up information and confidently state it as fact. GenAIs frequently produce code that does not run, is not optimal, and/or produces incorrect results. I'll talk about problems I've encountered using them since ChatGPT's release late in 2022.
- Your Python skills will be critical for building/proving models, and for evaluating/proving code written by genAIs produces correct results. Significant coding and software engineering skills will be required to guide genAIs into producing the best results.

Maximizing Your Learning Experience

We have a diverse audience of programming novices, intermediate programmers and a few advanced programmers. I'll work to maximize your learning experience regardless of your skill level. The lectures are aimed at programming novices to intermediate programmers to ensure you have a solid Python foundation on which you can build as you move through the MSBA program.

Questions & Answers

- During the Zoom lectures, please ask questions as you have them.
- Please email me with questions **any time** during and after the workshop. Send questions to paul@deitel.com and include "UCLA MSBA" in the subject line.

Communications

Check your email daily—email is critical for communication in industry:

- Professionals expect responses within one business day.
- I typically send one or two group emails per week with announcements or tips. You do not need to respond to these unless you have questions.
- If I write to you individually, please respond within one business day.
- If you email me, I'll respond within hours during the workshop or one business day after it. I am in the U.S. Eastern Time Zone, so many of you will email me in the middle of the night. I'll respond as soon as I can.

Grading

At the end of this workshop, you'll receive a course certificate signed by Paul Brandano, Assistant Dean of the Master of Science in Business Analytics program, and me. It will contain a grade of Completed, High Pass or Pass with Distinction. This grade will reflect your performance on the chapter quizzes and the required code-based homework lab exercises. This workshop grade does not count toward your MSBA GPA, but may be helpful as you interview for summer internships and full-time jobs. You must participate in the workshop and do the work to get a certificate.

About Me

I am CEO and CTO of Deitel & Associates, Inc. and an MIT graduate with 46+ years in computing. I am one of the world's most experienced programming-languages trainers, having taught worldwide in person and virtually to software developers of all skill levels at hundreds of Deitel & Associates, Inc.'s academic, industry, government and military clients since 1992. This is my 11th UCLA Anderson School of Management Python workshop - the last seven years for the MSBA program and the last four for the MSBA program. I am a top author on O'Reilly Online Learning, where my Python Fundamentals LiveLessons, Java Fundamentals LiveLessons, C++20 Fundamentals and C# Fundamentals 50+ hour video courses have ranked #1 ten of the last 16 years among thousands of video products from 200+ publishers - at least one of my courses has been in the top two or three nearly all 16 years. Tens of thousands of users have viewed my asynchronous streaming Python Fundamentals LiveLessons videos and attended my live Python and Python Data Science one-day, code-intensive webinars. My co-author, Dr. Harvey M. Deitel, and I are among the world's bestselling programming-language textbook and professional book authors.

Course Summary and Policies

Three-Hour Live Zoom Lectures

- 8:00AM - 11:00AM PDT with a 10-15 minute break in the middle.
- Dates: September 3, 10 & 24; October 8 & 22; November 5 & 12. Paul Brandano checks attendance in these sessions. If you have a conflict, please email paul.brandano@anderson.ucla.edu and CC paul@deitel.com.
- Presentations of chapter topics, code, demos, tips & tricks, Q&A, lab exercise review.
- Recorded - I'll post a link to the recorded lecture after Zoom completes the processing. Typically 3-4 hours after lecture.
- Please attend the live lectures - Rare Zoom failures might corrupt the recordings.
- Q&A outside of class via paul@deitel.com

Required Weekly Hands-On Lab Exercises

- Due Mondays at noon (12PM) sharp Pacific time.
- Submit to me as one fully executed Jupyter Notebook, which you'll upload in Bruin Learn.
- LabAssist notes: In the syllabus, I provide hints & exercise modifications to help you solve the exercises efficiently and effectively. Be sure to read them! I often modify what the original exercise asks.
- Questions? Send e-mail to paul@deitel.com
- I provide detailed comments/suggestions on your assignments. Precision and improvement are important - please read my comments carefully.

Chapter Assessment Quizzes in Bruin Learn (Chapters 2-10)

- One 9-or-10-question multiple-choice quiz per chapter.
- Due at 11:59PM (23:59) Pacific Time the Tuesday after lecture.
- Quizzes reward careful reading of the textbook.
- Must work on your own, but open book is fine.

Generative AI (GenAI) and Academic Integrity Policies

- Guiding Principles for Responsible Use: https://dts.ucla.edu/initiatives/ai/guiding-principles-responsible-use
- UCLA AI Website: https://dts.ucla.edu/initiatives/ai
- Student Conduct Code: https://studentconduct.ucla.edu/individual-student-code
- Using Generative AI Reflectively and Responsibly in Teaching and Learning: https://teaching.ucla.edu/resources/teaching-guides/using-generative-ai-reflectively-and-responsibly-in-teaching-and-learning/
- Python is crucial to your success in the MSBA program and career. Do the work to learn the language. **Without knowing Python, how would you determine if AI-generated code is correct and its outputs are valid?**
- You may use genAIs for research and learning purposes only.
- GenAIs "hallucinate": Verify everything they return. Learn how to leverage these tools effectively while maintaining a cautious and critical mindset.
- You may not use generative AI to solve coding exercises, explain your solutions or determine the answers to the chapter quizzes. You must submit your own work.
- Prompt engineering is a skill - the better your prompts, the better the genAI results.

Required Textbook

Intro to Python for Computer Science and Data Science: Learning to Program with AI, Big Data and the Cloud by Paul Deitel & Harvey Deitel, published by Pearson Higher Education

It's important to use the U.S. Edition, as the publisher makes changes to the international editions.

There are **rental and purchase options** for the e-book (immediately accessible) and print versions. I recommend the e-books as they're text-searchable, and you can highlight and take notes. **PLEASE DO NOT DOWNLOAD PIRATED ILLEGAL COPIES OF MY BOOK. It's against UCLA's code of conduct, and the pirated copies are loaded with errors that make it obvious to me when students are using them.**

E-Books

The e-books are available from various vendors and rental/purchase options vary in some countries:

- Pearson+ rental: https://deitel.com/IntroToPythonPearsonPlus
- VitalSource.com purchase: https://deitel.com/PYCDSVitalSource
- Redshelf.com purchase: https://deitel.com/PYCDSRedshelfPurchase (it appears there are purchase and rental options here now—at least in the U.S.)

Please do not purchase our professional book **Python for Programmers** - it does not have the end-of-chapter exercises used for this workshop's weekly required lab exercises.

Using the Textbook

I'll cover most concepts in the live Zoom lectures. The book contains additional insights and details not covered in the videos. The quizzes are based on the chapter content, so I recommend reading the chapters to ensure you do not miss details covered in the quizzes.

The Textbook's Examples, Exercises and Projects

- In this workshop, you'll use a hands-on applied approach to learn from a broad selection of real-world code examples and exercises drawn from computer science, data science and other fields. These are designed to give you an engaging, challenging and entertaining introduction to Python and Python data science.
- The book's Python code examples range from individual snippets to complete computer science, data science and AI case study code implementations. You should run every code example as you read the corresponding text. **I recommend typing the code yourself rather than just running the downloaded code. This helps you learn Python faster.**
- The Self-Check exercises and answers at the ends of most textbook sections enable you to immediately test your understanding of the concepts. The solution immediately follows each. After reading each section, try the Self-Check exercises and check your answers.
- The required hands-on labs are selected from the end-of-chapter exercises that complement the chapter code examples. Each chapter has many optional more substantial projects that encourage you to go deeper into what you've learned. If you already know Python, you should attempt some of these projects. Send me an email if you have questions.

Pre-Workshop Assignment to Complete Immediately

No submission for this assignment. You are simply getting your system ready for class.

Software Setup, Overview and Test-Drives

Please perform the following steps by August 31, so we can resolve any issues before the class begins September 3.

For errors, give them to a genAI chatbot and say what you were trying to do. They can probably help. If you still have issues, e-mail me at paul@deitel.com.

Install the Software

Download and install the Anaconda Distribution for Windows, macOS or Linux:

> https://www.anaconda.com/download/success

The site should detect your OS and provide an appropriate **Download** button. Double-click the installer file in your `Downloads` folder and follow the on-screen prompts. I use the default settings.

Install the Book's Code Examples

Get the book's code examples from

> https://github.com/pdeitel/IntroToPython

Clone the repository or click the green **<>Code** button and select **Download ZIP**. To extract the ZIP file's contents:

- Windows: Right-click the ZIP file in your `Downloads` folder, select **Extract All...**, then click **Extract**.
- macOS: Double-click the ZIP file in your `Downloads` folder.

Whether you use `git` tools to copy the repository or simply download the ZIP and extract its contents, GitHub will name the root folder `IntroToPython-master`. Within this folder is the subfolder named `examples`. Move this `examples` subfolder into your `Documents` folder. My instructions assume the examples reside in this folder.

Required Reading Assignments Before the September 3 Lecture

- Read the textbook's Preface for the book's scope, conventions and coverage.
- Work through Section 1.10 - Test-Drive: Using IPython and Jupyter Notebooks - to ensure you're familiar with IPython and Jupyter Notebooks.
- For more JupyterLab details, see **Getting Started** at

  > https://jupyterlab.readthedocs.io/en/stable/

Required Hands-On Lab Exercises

- Perform Section 1.10's test-drive steps.
- It's critical that your software is set up correctly before the workshop. If you encounter any software issues, contact me at paul@deitel.com so I can help. **Please include "UCLA MSBA Software Issue" in the subject.**

Schedule Overview — All Times Are Pacific Time (PDT)

Read before lecture to get more from this workshop.

Lecture 1, September 3

- Chapter 2, Intro to Python Programming.
- Lab exercises: Due Monday, September 7 at noon (12:00).
- Chapter 2 Quiz: Due Tuesday, September 8 at 11:59PM (23:59).

Lecture 2, September 10

- Finish Chapter 3, Control Statements and Program Development.
- If time: Start Chapter 4, Functions.
- Lab exercises: Due Monday, September 14 at noon (12:00).
- Chapter 3 Quiz: Due Tuesday, September 15 at 11:59PM (23:59).

Lecture 3, September 24

- Finish Chapter 4, Functions.
- Chapter 5, Lists and Tuples.
- Lab exercises: Due Monday, September 28 at noon (12:00).
- Chapters 4 & 5 Quizzes: Due Tuesday, September 29 at 11:59PM (23:59).

Lecture 4, October 8

- Chapter 6, Dictionaries and Sets.
- Chapter 7, Array-Oriented Programming with NumPy, Sections 7.1-7.13.
- Lab exercises: Due Monday, October 12 at noon (12:00).
- Chapter 6 Quiz: Due Tuesday, October 13 at 11:59PM (23:59).

Lecture 5, October 22

- Chapter 7, Array-Oriented Programming with NumPy, Section 7.14 (Intro to Pandas).
- Chapter 8, Strings: A Deeper Look (Intro to Pandas Data Wrangling).
- Lab exercises: Due Monday, October 26 at noon (12:00).
- Chapters 7 & 8 Quizzes: Due Tuesday, October 27 at 11:59PM (23:59).

Lecture 6, November 5

- Chapter 9, Files and Exceptions, Loading/Analyzing CSV Data with Pandas.
- Selections from Chapter 10, Object-Oriented Programming.
- Lab exercises: Due Monday, November 9 at noon (12:00).
- Chapters 9 & 10 Quizzes: Due Tuesday, November 10 at 11:59PM (23:59).

Lecture 7, November 12

- **NEW:** OpenAI's Python SDK, Agents SDK and Codex.
- OpenAI APIs and Agents SDK are paid services, **so this week's lab exercises are optional**. If you're interested in experimenting with the capabilities I demonstrate, you can do so for just a few dollars.

Exercise Requirements for All Weeks

Jupyter Notebook File Names

Upload to the Bruin Learn site one fully-executed Jupyter Notebook for each week's exercises. Name your notebook files as follows, replacing LastName, FirstName and StudentID with your information:

- LastName_FirstName_StudentID_Lecture1LabExercises.ipynb
- LastName_FirstName_StudentID_Lecture2LabExercises.ipynb
- LastName_FirstName_StudentID_Lecture3LabExercises.ipynb
- LastName_FirstName_StudentID_Lecture4LabExercises.ipynb
- LastName_FirstName_StudentID_Lecture5LabExercises.ipynb
- LastName_FirstName_StudentID_Lecture6LabExercises.ipynb

This helps me keep your homework files organized.

Author Your Notebooks as if They're for a Boss at Work

Structure the notebooks as if you're presenting data analytics results at your job. Notebooks use Markdown to describe how Jupyter should render text content as HTML. You place text-based markdown content in markdown cells and Python code in executable code cells. I'll discuss markdown basics in class.

Here is a nice markdown reference: https://daringfireball.net/projects/markdown/

Include a Self-Assessment Discussing Your Understanding of the Material

In a markdown cell at the top of your notebook, write 1-3 sentences telling me your thoughts on whether you understood the material, problems you had and, if you used AI for anything, how you used it. Examples:

- "I did all the required exercises and fully understand them."
- "I tried exercise X, but could not get it working. I am having trouble with ..." If you need extra help, do not wait until the last minute. I prefer that you reach out to me for help before submitting your assignment, rather than submitting an incomplete assignment. I am here to help you learn, so feel free to ask questions!

Notebook Format After the Self-Assessment

Include a level 1 header (# in markdown, also called an H1 in HTML) containing your notebook title, as in

```
# Lecture 1 Homework Exercises
```

and, for each homework problem, a level 2 header (## in markdown, also called an H2 in HTML) containing the exercise number and title from the syllabus, as in:

```
## 2.5 Circle Area, Diameter and CircuMSBArence
```

Follow each level 2 header with your code explanations in markdown cells and code in code cells. My lecture notebooks are a great reference for how to format your homeworks.

Required: You Must Explain Your Code

Communicating results will be a key aspect of your work in the MSBA program. Being able to explain what your code does and how it does it is a great way to prove to yourself and to me that you know the material. Use tools like the free version of Grammarly (https://grammarly.com) to eliminate spelling errors and ensure proper grammar. Formatting is a key part of communicating your explanations. I will give you a Jupyter Notebook containing a markdown reference. You can double click the formatted items in my notebook to see the corresponding markdown so you can learn how to do the formatting yourself. Google Colab supports Jupyter

Notebooks and provides a simple Markdown formatting bar with buttons like you might find in Microsoft Word or Google Docs. **Generative AIs are really good at Markdown—you may use them to help with that. Markdown formatting is also a key aspect of prompting generative AIs and defining skills for AI agents.**

Precision Is Critical

Read each exercise's description in the textbook then read its LabAssist notes in this syllabus. The LabAssist notes provide hints to help you work more efficiently and sometimes modify the exercise description or the approach you should use to solve each exercise. **Please do exactly what the combination of each exercise statement and its LabAssist notes ask.** Note that sometimes the LabAssist notes override parts of the exercise descriptions.

I am checking that you understand specific Python concepts. **You must use only the techniques that have been presented through the chapter in which the exercise appears.** For example, to solve exercises in Chapter 3, you must use techniques only from Chapters 1-3.

Lecture 1 - Thursday, September 3, 2026

Python Programming Fundamentals

Lecture: 8:00AM - 11:00AM Pacific Time (PDT)

- Q&A regarding IPython and Jupyter Notebooks.
- IPython and Jupyter Notebooks tips and tricks.
- Textbook: Chapter 2, Introduction to Python Programming.
- Chapter 2 Lab Exercises: Due Monday, September 7, noon (12:00) Pacific Time (PDT)
- Chapter 2 Quiz: Due Tuesday, September 8, 11:59PM (23:59) Pacific Time (PDT)

Upload to Bruin Learn a single, fully executed Jupyter Notebook containing your solutions to all the required Chapter 2 lab exercises. **Name your notebook** `LastName_FirstName_StudentID_Lecture1LabExercises.ipynb`

Include your self-assessment at the top of the notebook. NOTE: If the exercise numbers and titles in your book do not match the ones shown in this syllabus, then you do not have the U.S. edition. **REMINDER:** Explain every code solution in properly formatted markdown cells. **Use only techniques presented in Chapter 2.**

- **2.5 Circle Area, Diameter and CircuMSBArence** - **LabAssist:** See the exercise description for the formulas to use.
- **2.6 Odd or Even** - **LabAssist:** Input an integer from the user then test whether it's odd or even - see the hint in the exercise on how to do this. Recall that the `input` function returns a string, so you must convert the user inputs to integers. Thoroughly test your code - for this exercise you should duplicate your code in multiple notebook cells so you can test both odd and even numbers to ensure your code works correctly. (Some of you are likely familiar with Python's `if...else` statement. **Do not use it here as it has not yet been presented.**)
- **2.12 7% Investment Return** - **LabAssist:** See the exercise description for the formula to use. Your code should define the variables listed in the exercise description and use them in the formula, rather than simply inserting the numeric values in the formula. (Some of you are likely familiar with Python's looping statements. **Do not use them here as they have not yet been presented.**)
- **2.13 How Big Can Python Integers Be?** - **LabAssist:** Use the `**` operator to raise 10 to the powers 10, 100 and 1000 to show this. Based on the results you see from these three expressions, provide a markdown cell containing your thoughts on the question in the exercise title.
- **Optional but recommended** - The `ch02.ipynb` notebook I gave you for the Week 1 lecture contains many Generative AI sections with prompts. I encourage you to try these with multiple genAIs - I use ChatGPT, Copilot, Gemini, Perplexity and Claude.

Lecture 2 - Thursday, September 10, 2026

Control Statements

Lecture: 8:00AM - 11:00AM Pacific Time (PDT)

- Q&A for Chapter 2 exercises.
- Textbook: Chapter 3, Control Statements and Program Development.
- Textbook: (if time) Start Chapter 4, Functions.
- Chapter 3 Lab Exercises: Due Monday, September 14, noon (12:00) Pacific Time (PDT)
- Chapter 3 Quiz: Due Tuesday, September 15, 11:59PM (23:59) Pacific Time (PDT)

Upload to Bruin Learn a single, fully executed Jupyter Notebook containing your solutions to all the required Chapter 3 lab exercises. **Name your notebook** `LastName_FirstName_StudentID_Lecture2LabExercises.ipynb`

Include your self-assessment at the top of the notebook. NOTE: If the exercise numbers and titles in your book do not match the ones shown in this syllabus, then you do not have the U.S. edition. **REMINDER:** Explain every code solution in properly formatted markdown cells. **Use only techniques presented in Chapters 2 and 3.**

- **3.10 7% Investment Return** - **LabAssist:** Use a `for` loop.
- **3.17 Nested Loops** - **LabAssist:** See the hint in the exercise description. This exercise is particularly useful for novices learning about nested loops. **To print each asterisk (`*`), you must use only the `print` statement shown in the exercise description.** Though the exercise description shows the triangles of asterisks side by side, that is not how you should display them for this exercise.
- **3.26 Data Science Research: Anscombe's Quartet** - **LabAssist:** A classic for every budding data scientist. This is a quick, but important, Internet research project. Check out the Anscombe's Quartet Wikipedia article then, in a markdown cell, provide 2 to 4 sentences on what you learned.
- **3.31 Data Science: Categorical Data** - **LabAssist:** This is a thought exercise on which measures of central tendency are appropriate for categorical (non-numeric) data. Write your answer in a markdown cell. Focus specifically on the data and stats in the exercise description.
- **3.1 Validating User Input** - **LabAssist:** This exercise asks you to modify the script of Fig. 3.3 to validate its inputs. **Instead, prompt genAIs with the problem statement, asking them to modify the script and explain their changes.** Paste their code into markdown cells and test it thoroughly. In a markdown cell, explain whether your initial prompt worked and, if not, how you tuned your prompt to get the correct result. Also, compare the code they generated.
- **Optional but recommended** - The `ch03.ipynb` notebook I gave you contains many Generative AI sections with prompts. I encourage you to try these with multiple genAIs - I use ChatGPT, Copilot, Gemini, Perplexity and Claude.

Lecture 3 - Thursday, September 24, 2026

Chapter 4, Functions and Chapter 5, Lists and Tuples

Lecture: 8:00AM - 11:00AM Pacific Time (PDT)

- Q&A for Chapter 3 exercises.
- Textbook: Finish Chapter 4, Functions
- Textbook: Chapter 5, Lists and Tuples
- Chapters 4 & 5 Lab Exercises: Due Monday, September 28, noon (12:00) Pacific Time (PDT)
- Chapters 4 & 5 Quizzes: Due Tuesday, September 29, 11:59PM (23:59) Pacific Time (PDT) – **Note that there are two quizzes this week.**

Upload to Bruin Learn a single, fully executed Jupyter Notebook containing your solutions to all the required Chapter 4 & 5 lab exercises. **Name your notebook** `LastName_FirstName_StudentID_Lecture3LabExercises.ipynb`

Include your self-assessment at the top of the notebook. NOTE: If the exercise numbers and titles in your book do not match the ones shown in this syllabus, then you do not have the U.S. edition. **REMINDER:** Explain every code solution in properly formatted markdown cells. **Use only techniques presented in Chapters 2 through 4 or 5, depending on the exercise.**

- **4.9 Data Science: Temperature Conversion** - **LabAssist:** Conversion is a common data-wrangling operation.
- **5.6 Functions Returning Tuples** - **LabAssist:** The function you'll write for this exercise uses tuple packing and unpacking, similar to how we swapped two variables' values in Section 5.4.
- **5.7 Data Science: Duplicate Elimination** - **LabAssist:** Duplicate elimination is a data-wrangling operation. For each item in the original list, use not in to determine whether the item is in the new list. If not, add it. **Do not use NumPy's `unique` function in your solution.**
- **5.14 Is a Sequence Sorted?** - **LabAssist:** One way to check this is to create a sorted copy of the sequence using built-in function **sorted**, then compare the copy to the original. If they're the same, the original sequence is sorted. **Do not do this!** Sorting is an "expensive" operation, especially as the number of elements to be sorted increases. In big data, it's common to process enormous sequences with millions or even billions of elements, so making a copy and sorting it can be expensive - if you even have enough memory available to do that. Sometimes simple solutions require large amounts of processor time and memory. Solutions that minimize processor time can require more memory, and solutions that minimize memory consumption can require more processor time. This is called the space-time trade-off. [See Chapter 11 for a nice intro to computer-science thinking.] A more efficient approach to this exercise is to make one pass of the sequence. Compare element 0 with element 1 to determine whether they're in ascending order. If so, compare element 1 and element 2 to determine whether they're in ascending order. Continue these comparisons with elements 2 and 3, 3 and 4, etc. until you encounter adjacent elements that are not in ascending order (in which case, the sequence is not sorted) or until you reach the end of the sequence (in which case, it is sorted). **Allow duplicate values.**
- **5.28 Data Science: Survey Response Statistics** - **LabAssist:** Recall that you must pass a second argument to NumPy's `unique` function to indicate that it should count the frequencies of the unique values - see the code in Section 5.17.2.

Lecture 4: Thursday, October 8, 2026

Chapter 6, Dictionaries and Sets and Chapter 7, Array-Oriented Programming with NumPy

Lecture: 8:00AM - 11:00AM Pacific Time (PDT)

- Q&A for Chapters 4 and 5 exercises.
- Textbook: Chapter 6, Dictionaries and Sets
- Textbook: Chapter 7, Array-Oriented Programming with NumPy (sections 7.1-7.13).
- Chapters 6 & 7 Lab Exercises: Due Monday, October 12, noon (12:00) Pacific Time (PDT)
- Chapter 6 Quiz: Due Tuesday, October 13, 11:59PM (23:59) Pacific Time (PDT)

Upload to Bruin Learn a single, fully executed Jupyter Notebook containing your solutions to all the required Chapter 6 & 7 lab exercises. **Name your notebook** `LastName_FirstName_StudentID_Lecture4LabExercises.ipynb`

Include your self-assessment at the top of the notebook. NOTE: If the exercise numbers and titles in your book do not match the ones shown in this syllabus, then you do not have the U.S. edition. **REMINDER:** Explain every code solution in properly formatted markdown cells. Use only techniques presented in Chapters 2 through 6 or 7, depending on the exercise.

- **6.6 Data Science: Duplicate Word Removal** - **Use the following updated version of the exercise description, NOT the one in the book:**
  (Duplicate Word Removal) Write a function that receives a list of words, then determines and returns a sorted list of the unique words. Treat uppercase and lowercase the same - for example, if the list contains `apple` and `Apple`, the list you return should contain `apple` once. The **caller** should display the unique words in alphabetical order with one word per line. Test your function by calling it several times with different sentences. Remember to break each sentence into a list of words **before** passing it to the function. **LabAssist:** The techniques used here are data-wrangling operations. In Chapter 5, you learned that built-in function `sorted` sorts a sequence of values and returns a new sorted list. Recall that string sort order is determined lexicographically - that is, using the characters' underlying integer values. To sort strings alphabetically, treat all their letters as the same case, as demonstrated in Section 5.15.
- **6.9 Dictionary Manipulations** - **LabAssist:** If necessary, see the dictionary operations demonstrated in Section 6.2 for similar operations.
- **7.2 Broadcasting** - **LabAssist:** Model your code after the broadcasting examples in Section 7.7.
- **7.9 Data Science: Indexing and Slicing `array`s** - **LabAssist:** Model your code after the examples in Section 7.10. These techniques can be used to select portions of datasets for analysis. **Part (b) has a typo in the original exercise description - it should say, "Select column 4." not "Select column 5."**

Lecture 5: Thursday, October 22, 2026

Chapter 7, Array-Oriented Programming with NumPy - Intro to Pandas Part 1, and Chapter 8, Strings: A Deeper Look and an Intro to Pandas Part 2

Lecture: 8:00AM - 11:00AM Pacific Time (PDT)

- Q&A for Chapters 6 and 7 exercises.
- Textbook: Chapter 7, Array-Oriented Programming with NumPy, Section 7.14
- Textbook: Chapter 8, Strings: A Deeper Look
- Chapters 7 & 8 Lab Exercises: Due Monday, October 26, noon (12:00) Pacific Time (PDT)
- Chapters 7 & 8 Quizzes: Due Tuesday, October 27, 11:59PM (23:59) Pacific Time (PDT)

Upload to Bruin Learn a single, fully executed Jupyter Notebook containing your solutions to all the required Chapter 7 & 8 lab exercises. **Name your notebook** `LastName_FirstName_StudentID_Lecture5LabExercises.ipynb`

Include your self-assessment at the top of the notebook. NOTE: If the exercise numbers and titles in your book do not match the ones shown in this syllabus, then you do not have the U.S. edition. **REMINDER:** Explain every code solution in properly formatted markdown cells. **Use only techniques presented in Chapters 2 through 7 or 8, depending on the exercise.**

- **7.23 Data Science: Pandas DataFrames** - **LabAssist:** Model your code after the code examples in Section 7.14.2.
- **8.3 (Pig Latin)** - **LabAssist:** Remember that strings can be sliced like lists and arrays. Ensure that you account for all the rules the exercise describes for conversions.
- **8.8 Converting Integers to Emojis** - **LabAssist:** For fun, go to https://unicode.org/emoji/charts/full-emoji-list.html research the ranges of Unicode values for all emoji characters and try to print them all. Does your system support all of them?

Lecture 6: Thursday, November 5, 2026

Chapter 9, Files and Exceptions and Intro to Pandas, Part 3, and Selections from Chapter 10, Object-Oriented Programming

Lecture: 8:00AM - 11:00AM Pacific Time (PDT)

- Q&A for Chapters 7 and 8 exercises.
- Textbook: Chapter 9, Files and Exceptions
- Textbook: Chapter 10, Object-Oriented Programming, Sections 10.1-10.4, 10.7-10.9, Section 10.17.
- Chapters 9 & 10 Lab Exercises: Due Monday, November 9, noon (12:00) Pacific Time (PDT)
- Chapters 9 & 10 Quizzes: Due Tuesday, November 10, 11:59PM (23:59) Pacific Time (PDT)

Upload to Bruin Learn a single, fully executed Jupyter Notebook containing your solutions to all the required Chapter 9 & 10 lab exercises. **Name your notebook** `LastName_FirstName_StudentID_Lecture6LabExercises.ipynb`

Include your self-assessment at the top of the notebook. NOTE: If the exercise numbers and titles in your book do not match the ones shown in this syllabus, then you do not have the U.S. edition. **REMINDER:** Explain every code solution in properly formatted markdown cells. **Use only techniques presented in Chapters 2 through 9 or 10, depending on the exercise.**

- **9.16 Data Science: Working with the `diamonds.csv` Dataset in Pandas** - **LabAssist:** Model your code after the corresponding code snippets for the Titanic Disaster dataset presented in Section 9.12.2.
- **10.2 Account Class with Read-Only Properties** - **LabAssist:** Use the syntax for property getters shown in Section 10.4.2's `Time` class.
- **10.5 Duck Typing** - **LabAssist:** For class `SalariedEmployee`, you can mimic much of the syntax and code in class `CommissionEmployee`.

Lecture 7: Thursday, November 12, 2026

NEW: Programming with OpenAI's Python SDK, Agents SDK and Codex

Lecture: 8:00AM - 11:00AM Pacific Time (PDT)

- Q&A for Chapters 9 and 10 exercises.
- No reading assignment this week.

This lecture is under development now. I'll post a syllabus update once I complete it for the MFE program workshop I am currently teaching.

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